

Research: 網路診斷&容錯

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* System Level Fault Diagnosis.

* Background and Motivation.

現代多處理器 / 多電腦系統，使用各種拓撲的 interconnection network connect the node.

common topological example { Hypercube (3-cube)
Star Graph
k-ray n-cube (Q_2^4 , 4 ray 2 cube)
Bubble sort Graph (B_4)

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系統規模越大，node fault 越不可回避，For example: Single processer reliability = 99.9%
process number: 100, 1000, 10000, so the process, reliability are: 99.01%, 90.48%, 36.79%.
且加上 network attack is more and more., 所以要找 method for muti fault node in system.

* System Level Diagnosis.

使系統中 node 互相檢查 neighbor node status and find all faulty units

if node i is checker, but it is fault node, then the check is incorrect / untrustworthy.

* PMC model

Test outcome (a_{ij}) definition:

$$\begin{cases} i \text{ good} \rightarrow j \text{ good} : a_{ij} = 0 \\ i \text{ good} \rightarrow j \text{ bad} : a_{ij} = 1 \\ i \text{ bad} \rightarrow j (\text{bad or good}) : a_{ij} = \times (0/1, \text{untrustworthy}) \end{cases}$$

由一个 centralized arbiter 会根据 all result 判断 each process is faulty or faulty-free

* t -Diagnosability

对一个给定的 network, 存在一个关键数值 t , 只要 bad node not over the t arbiter 就能靠互测结果正确判断 all bad node. 此 t 称为该 network 的 t -Diagnosability

but the bad node number over the t , 此互测将失效



t -Diagnosability is 1



会与其他 syndrome 混淆, t -Diagnosable not 2 ..

* Conditional Diagnosability

原本定义的 t -diagnosability 对坏点分布没有任何限制

(坏点可无条件, 任意分布)

可以假设, 任一节点的所有 neighbor 不会同时全部故障

所以, 假如接受此假设, 可排除很多不正确的检测结果

arbiter 能容忍的 bad node 提升, t -Diagnosability 提高

, 而此假设下称为 Conditional Diagnosability